

HACKIA'26

XAI and GenAI for SmartHomes & Cities

Groupe 3

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THE SAFEAI VISION

Michel, a dedicated firefighter, faces a dual challenge: risking his life on the front lines and caring for his father, who lives with Alzheimer's.

Goal: Develop an AI system capable of assisting him both in high-risk field operations and ensuring daily safety at home.



PROJECT ARCHITECTURE



Firefighter Support-Phase 1

- Real-time Fire Detection
- Smoke & Flame Classification
- XAI-based Visual Justifications

Pauline, Harisson



Smart Home Safety-Phase 2/3

- Facial Recognition for Security
- Personal Object Tracking
- Fall Detection

Arnaud, Leonardo, Tristan

PHASE 1: FIRE DETECTION

Models

- **EfficientNet-V2-S** — compact modern CNN
 - chosen for: low latency, small footprint → Jetson target
- **Swin-B** — hierarchical Vision Transformer
 - chosen for: performance + global attention → better XAI
- Both pre-trained on ImageNet, fine-tuned with same recipe.

Dataset

- 3 classes: `no_fire`, `start_fire`, `fire`
 - Fusion: **DB3** (provided) + subset of **D-Fire train**
 - **6,000 images** total — **2,000 / class**
 - Split 70 / 15 / 15: train 4,200 / val 900 / test 900
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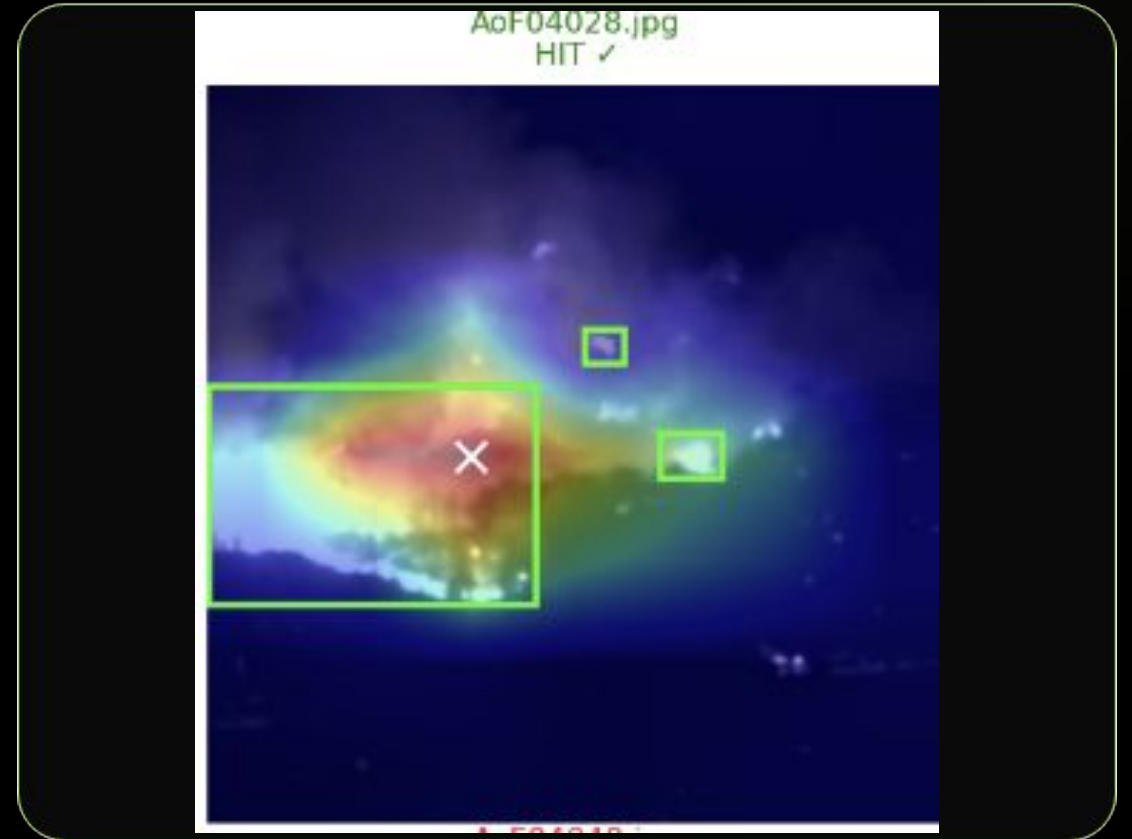
PHASE 1: Results

	Swin	EfficientNet
Test accuracy	92.22%	92.89%
Recall <code>fire</code>	97.67%	96.00%
Checkpoint	1042 MB	244 MB
Params	86.8 M	20.2 M
Latency FP16 (ms)	13.5	10.6
Throughput (img/s)	~74	~94

PHASE 1: Explainability (Pointing Game)

- Metric (Zhang et al. 2018): $\text{argmax}(\text{Grad-CAM})$ inside GT bbox = hit.
- Evaluated on D-Fire (3,445 images).
- Same accuracy $\neq$ same explainability.
- Transformer attention $\rightarrow$ better localization.

Model	PG <code>fire</code>	PG <code>smoke</code>
Swin-B	0.509	0.631
EffNet-V2-S	0.415	0.510



PHASE 1: FIRE DETECTION

Problem : specific images detected as fire or start_fire

-> Data augmentation

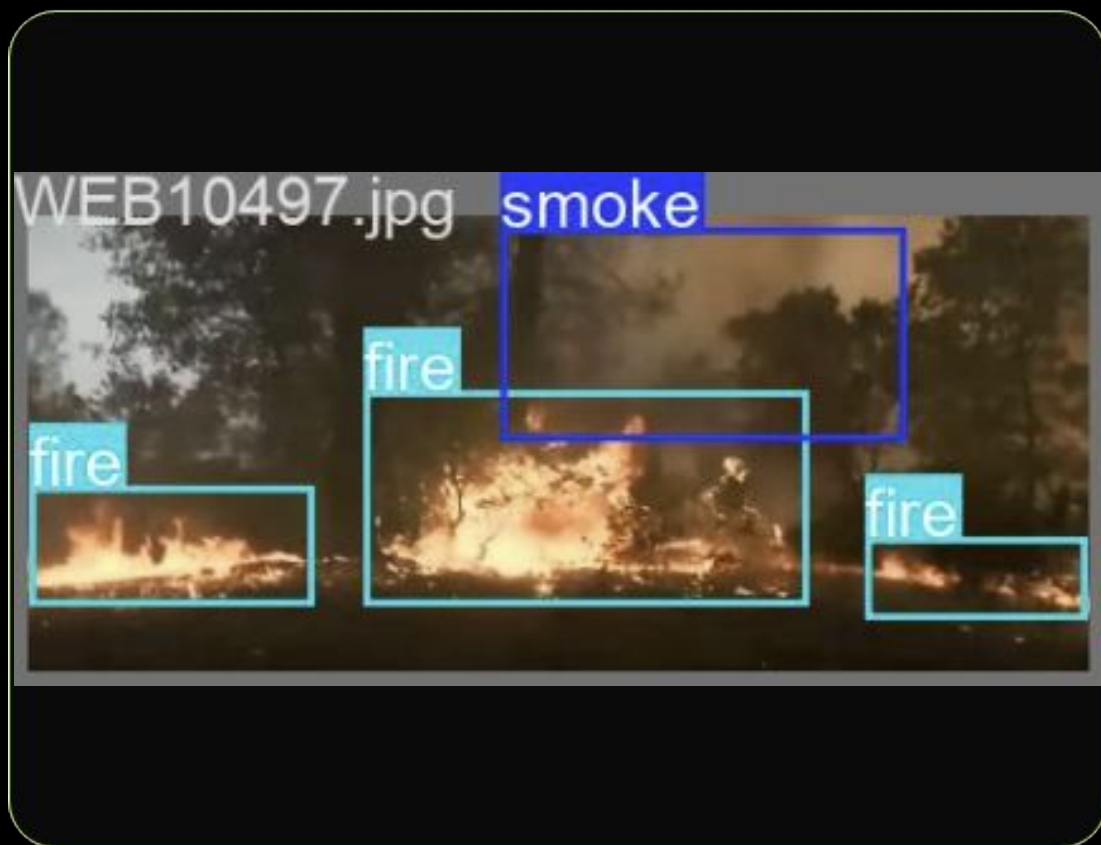
- Database 3 from the first challenge augmented using five different methods (horizontal flip, rotation, brightness and contrast adjustment, gaussian noise, gaussian blur)
- Around one hundred data entries using images and videos to address the different issues (fire truck, plane, flames)

-> around 3000 images for each classes

Trained model showing high accuracy on this database ($97.0\% \pm 0.9$)



PHASE 1: FIRE LOCALIZATION



Yolo26-m

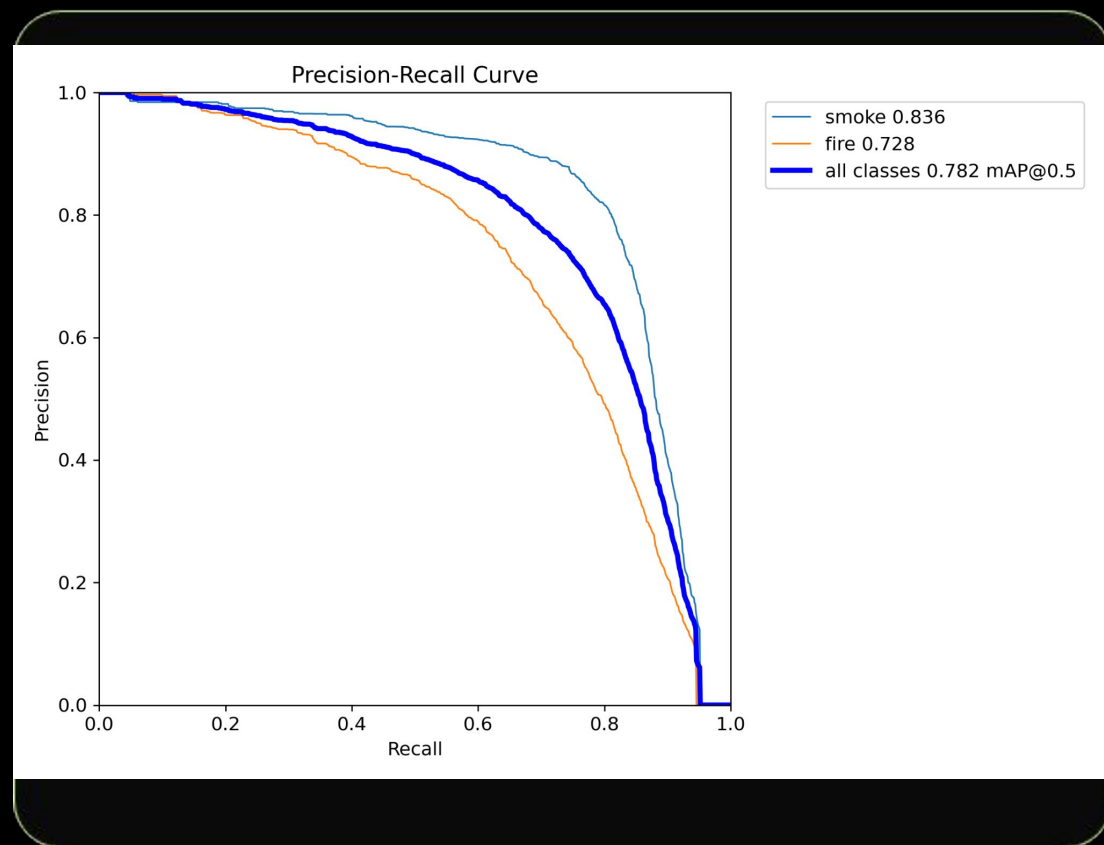
Good speed/accuracy tradeoff to fine-tune

Transfer Learning from COCO → re-init only the class heads (smoke & fire)

Dataset **D-Fire** — 13.8k train / 3.4k val / 4.3k test

Data Augmentation mosaic + mixup + copy-paste + erasing + RandAugment

PHASE 1: FIRE LOCALIZATION : Results



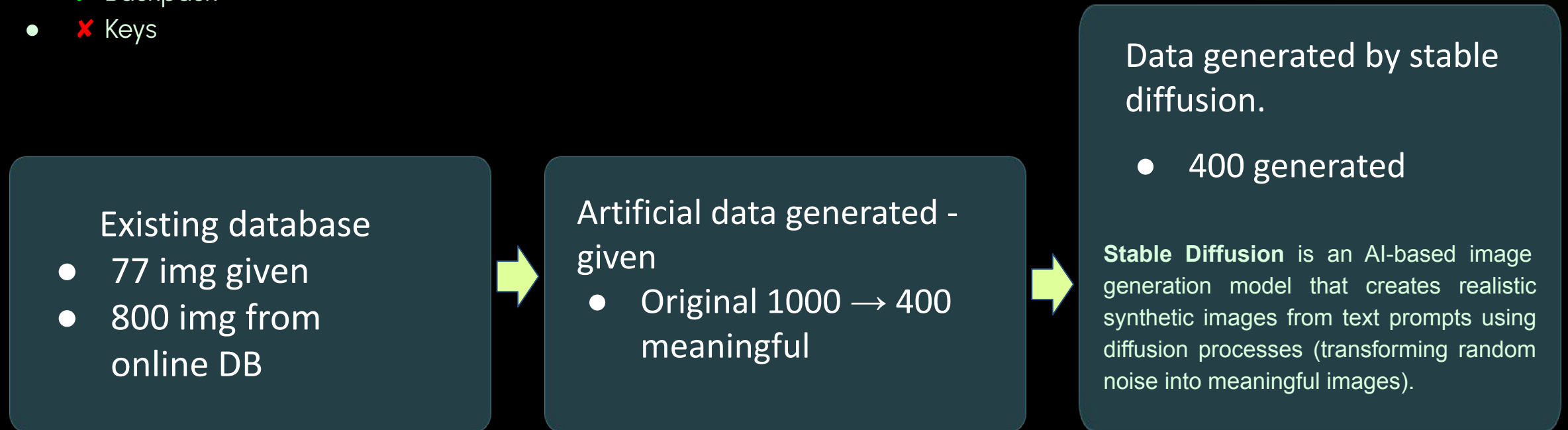
Classes	mAP50	mAP50-95
All	0.782	0.444
smoke	0.836	0.506
fire	0.728	0.381

PHASE 2: PERSONAL OBJECT TRACKING

YOLO 26n pretrained on COCO dataset (80 standard object classes)

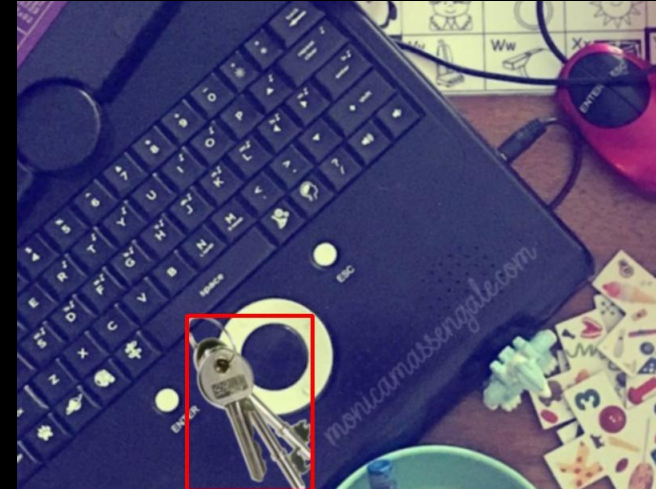
COCO pretrained classes:

- ✓ Smartphone
- ✓ Backpack
- ✗ Keys



✨ ARTIFICIAL GENERATED

- **Artificial generation:** Keys randomly overlaid on backgrounds (office, living room...).
- **Realism enhancement:** Gaussian blur, compression, adding noise.
- **Manual filtering:** Removal of anomalies (scale, context, contrast).
- **Tests conducted:** Comparison between the original dataset and the modified dataset.



✂️ DATA GENERATION BY STABLE DIFFUSION

Classical text to image stable diffusion + inpainting did not give good results for keys generation in given background because of their small and detailed geometry. Therefore, as suggested we used a different generation strategy.

- Background→Mask→Diffusion (key) **NOK**
- Mask(background) → Diffusion (background) **OK**

01 KEY ISOLATION

The key is isolated from a picture



02 MASK CREATION

A mask is created to protect the key (black=keep, white=regenerate)



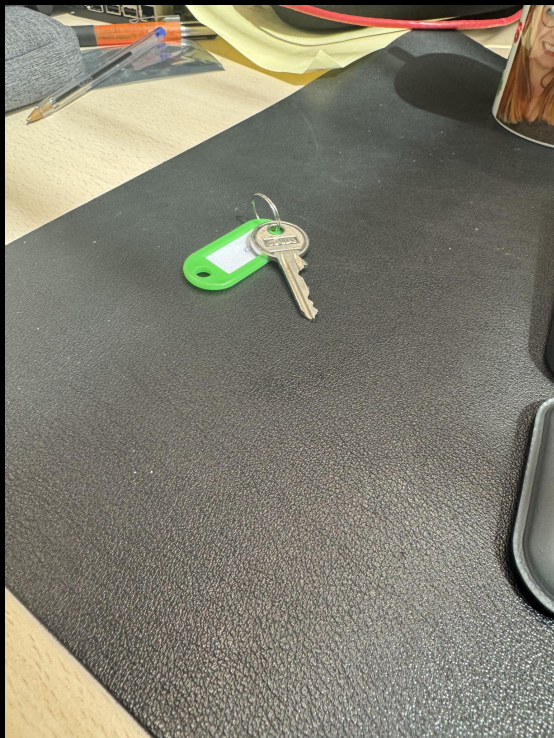
03 Diffusion inpainting

Stable diffusion inpainting regenerates the background while keeping the key fixed.

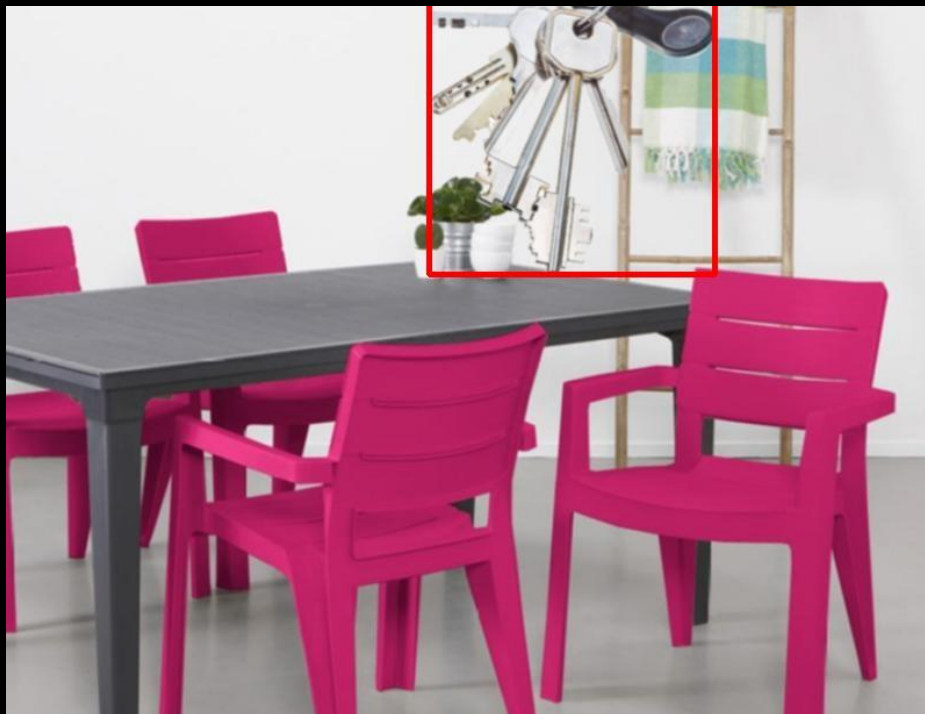


🌟✏️ Data Comparison

Real



Generated



Diffusion



|| EVALUATION PERFORMANCE MATRIX

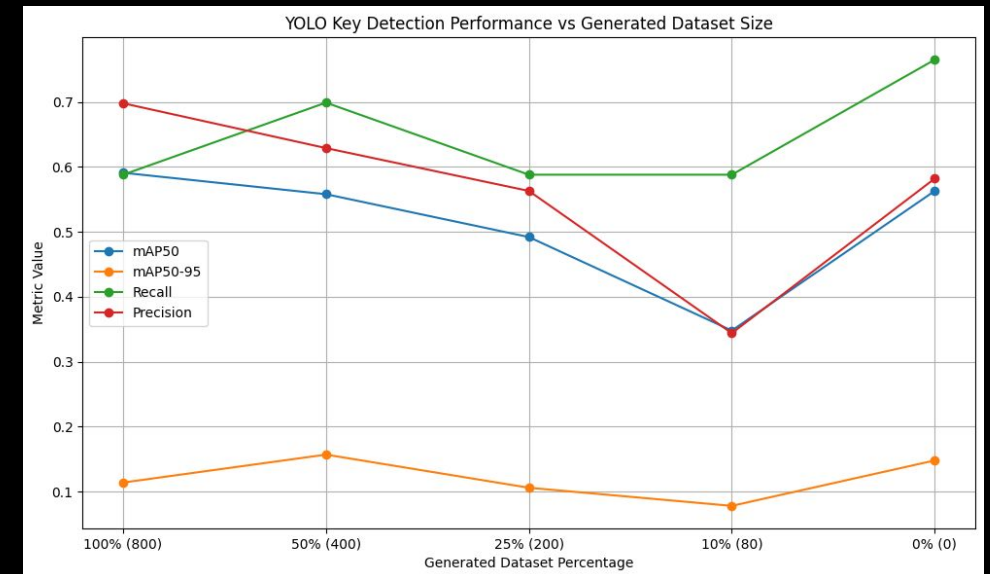
The provided generated dataset and the diffusion-based generated images showed qualitatively similar characteristics. Therefore, the objective was not to compare the two generation methods directly, but rather to evaluate the impact of adding generated data to the real dataset at different proportions.

Several training configurations were tested using combinations of:

- real annotated images
- generated images (artificial + diffusion)

The percentages indicate the ratio between generated and real data.

Dataset	mAP50	mAP50-95	Recall	Precision
100% (800)	0.591	0.114	0.588	0.698
50% (400)	0.558	0.157	0.699	0.629
25% (200)	0.492	0.106	0.588	0.563
10% (80)	0.348	0.078	0.588	0.344



PHASE 3: FALL DETECTION



Automated behavior analysis via computer vision to trigger instant alerts in case of an accident.

Model architecture : VideoMAE, a self-supervised Vision Transformer that learns by reconstructing masked video patches.

Deployment: Powered by a specialized HuggingFace model, optimized strictly for Fall / No Fall classification.

Successfully integrated this model into the interface and managed to achieve real-time fall detection.

| PHASE 4: Test of models with intuitive interface

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